INTRODUCTION TO THE DIGESTIVE SYSTEM

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DIKIOHS,DUHS.

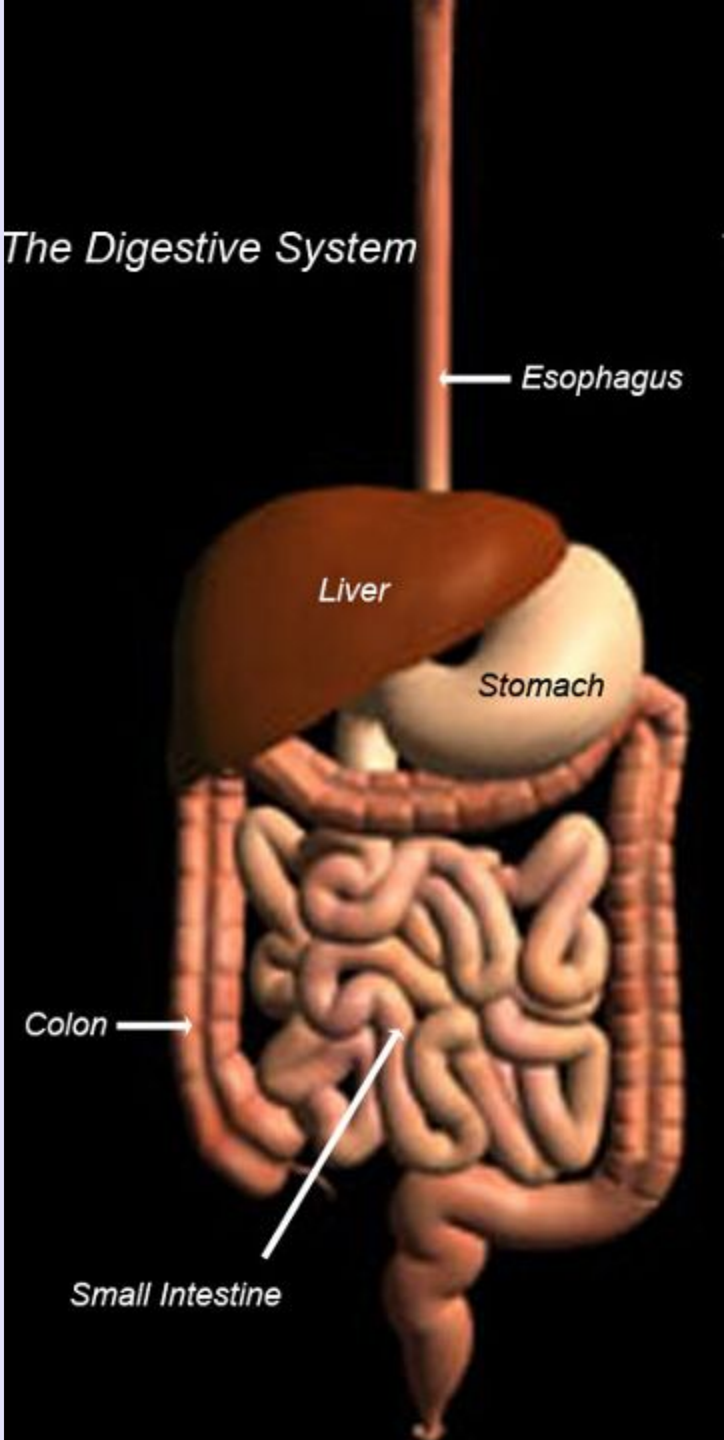
Learning Objectives

By the end of the lecture, the student should be able to**:**

- Understand the **functional significance** of the gastrointestinal system
- Describe the **structure** of the gastrointestinal tract, the glands that drain into it, and its **subdivision**.
- List the **major gastrointestinal secretions**.
- Identify the **major hormones** of the gastrointestinal system.

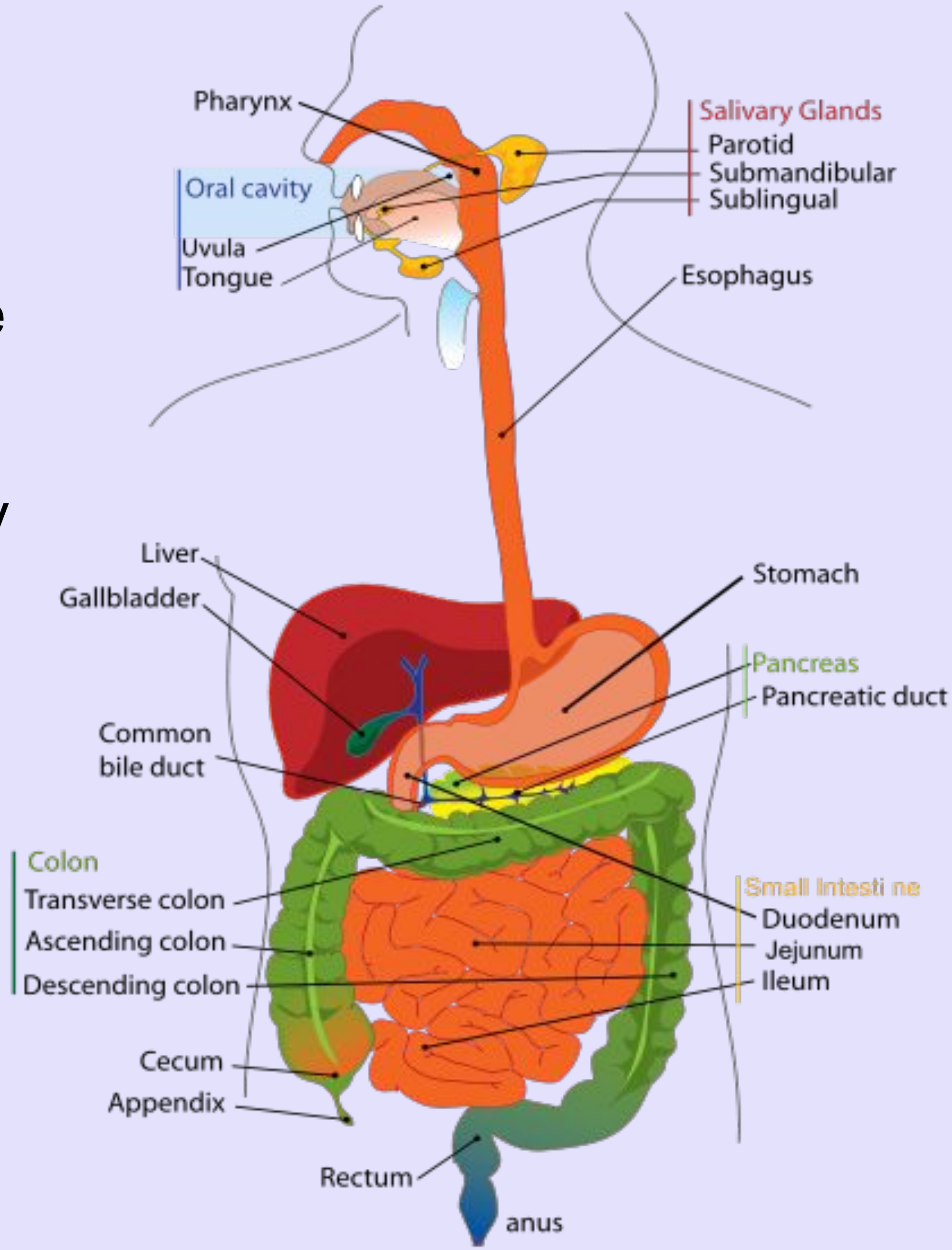
The Digestive System Introduction

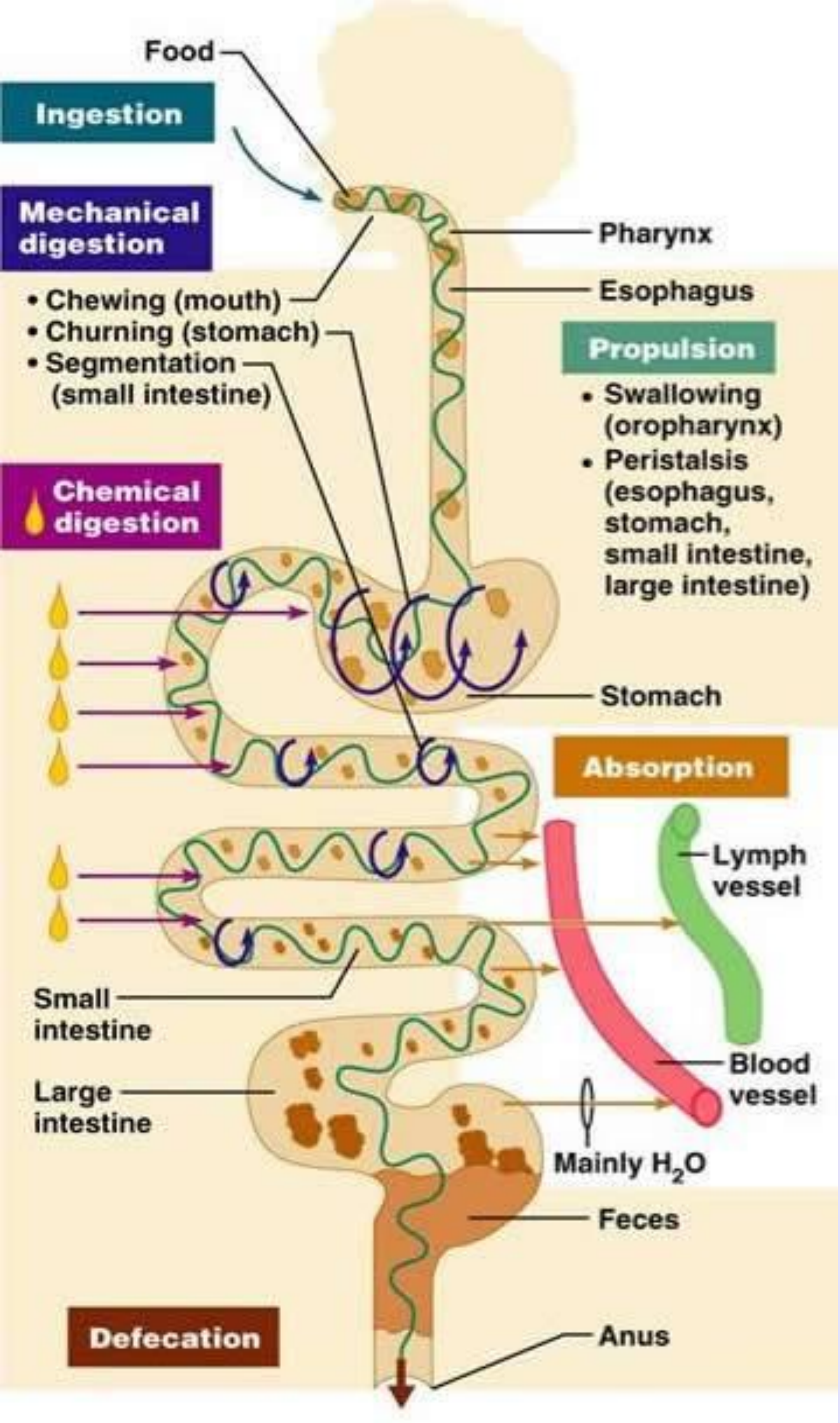
- The functions of the gastrointestinal tract are digestion and absorption of nutrients. To serve these functions, there are four major activities of the gastrointestinal tract.
- (1) Motility propels ingested food from the mouth toward the rectum and mixes and reduces the size of the food. The rate at which food is propelled through the gastrointestinal tract is regulated to optimize the time for digestion and absorption.
- (2) Secretions from the salivary glands, pancreas, and liver add fluid, electrolytes, enzymes, and mucus to the lumen of the gastrointestinal tract. These secretions further aid in digestion and absorption.
- (3) Ingested foods are digested into absorbable molecules.
- (4) Nutrients, electrolytes, and water are absorbed from the intestinal lumen into the bloodstream.



GIT

- The **gastrointestinal tract** is a continuous tube that stretches from mouth to the anus.
- Meal is mixed with a variety of secretions that arise from both the gastrointestinal tract itself and organs that drain into it, such as pancreas, gallbladder, and salivary glands.





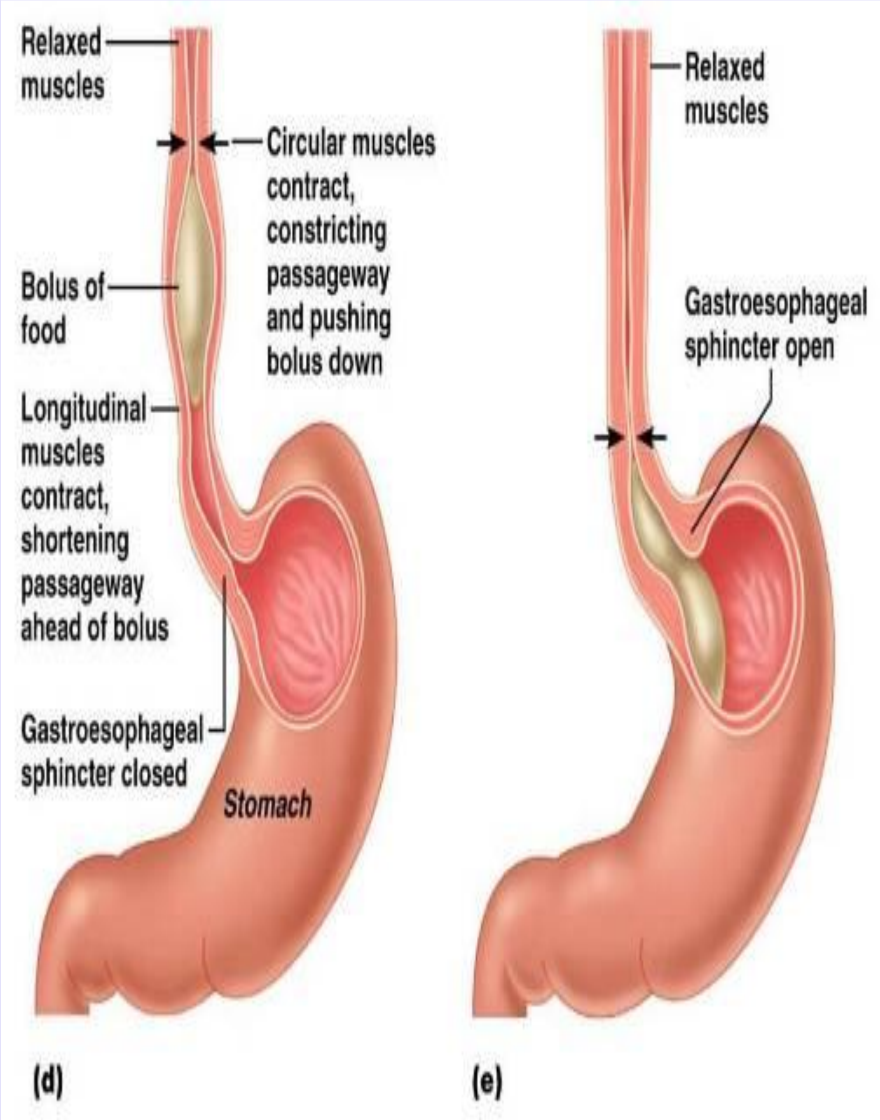
Basic Digestive Processes

The digestive system performs four basic digestive processes:

- Motility
- Secretion
- Digestion
- Absorption

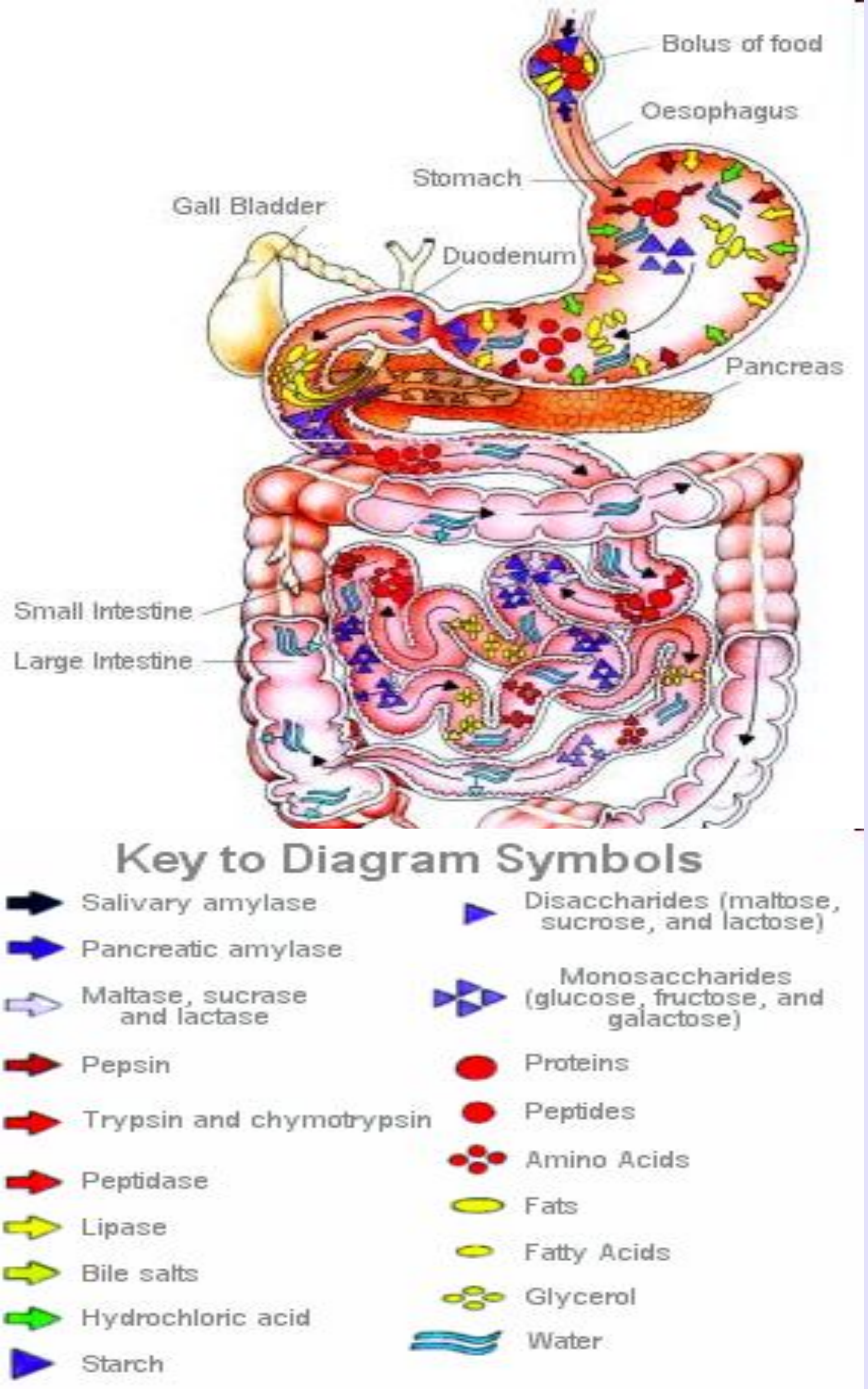
Motility

- Motility is a general term that refers to contraction and relaxation of the walls and sphincters of the gastrointestinal tract. Motility grinds, mixes, and fragments ingested food to prepare it for digestion and absorption, and then it propels the food along the gastrointestinal tract.
- Two basic types of digestive motility are superimposed on the ongoing tonic activity:
  - Propulsive Movements.
  - Mixing Movements.



Secretion

- Secretion is the addition of fluids, enzymes, and mucus to the lumen of the gastrointestinal tract. These secretions are produced by salivary glands (saliva), the cells of the gastric mucosa (gastric secretion), the exocrine cells of the pancreas (pancreatic secretion), and the liver (bile).
- Each digestive secretion consists of water, electrolytes, and specific organic constituents that are important in digestive processes such as enzymes, bile salts or mucous.

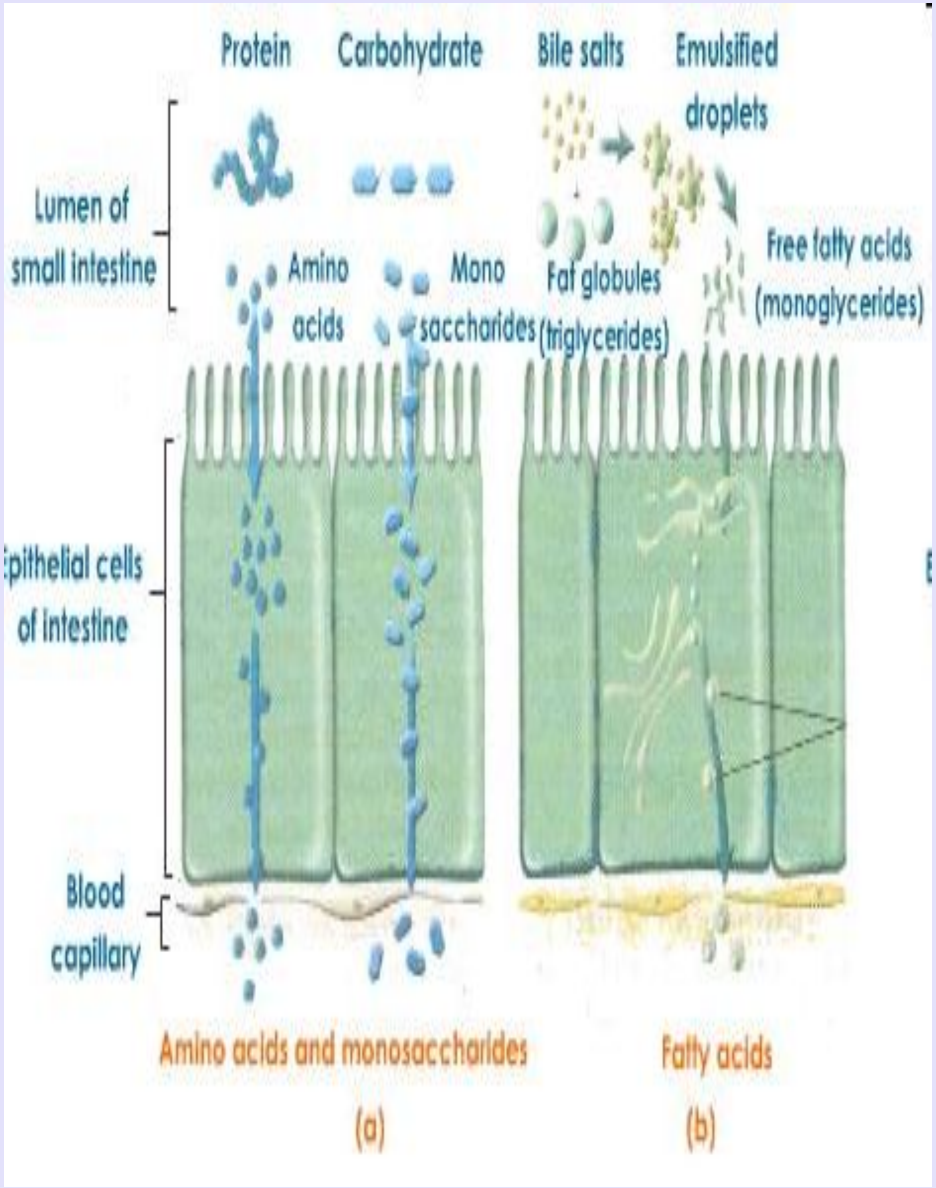


Digestion

- Biochemical breakdown of structurally complex foodstuffs (carbohydrates, proteins and fats) of the diet into smaller, absorbable units by enzymes produced within digestive system.
- Digestion is the chemical breakdown of ingested foods into absorbable molecules.
- The digestive enzymes are secreted in salivary, gastric, and pancreatic juices and also are present on the apical membrane of intestinal epithelial cells

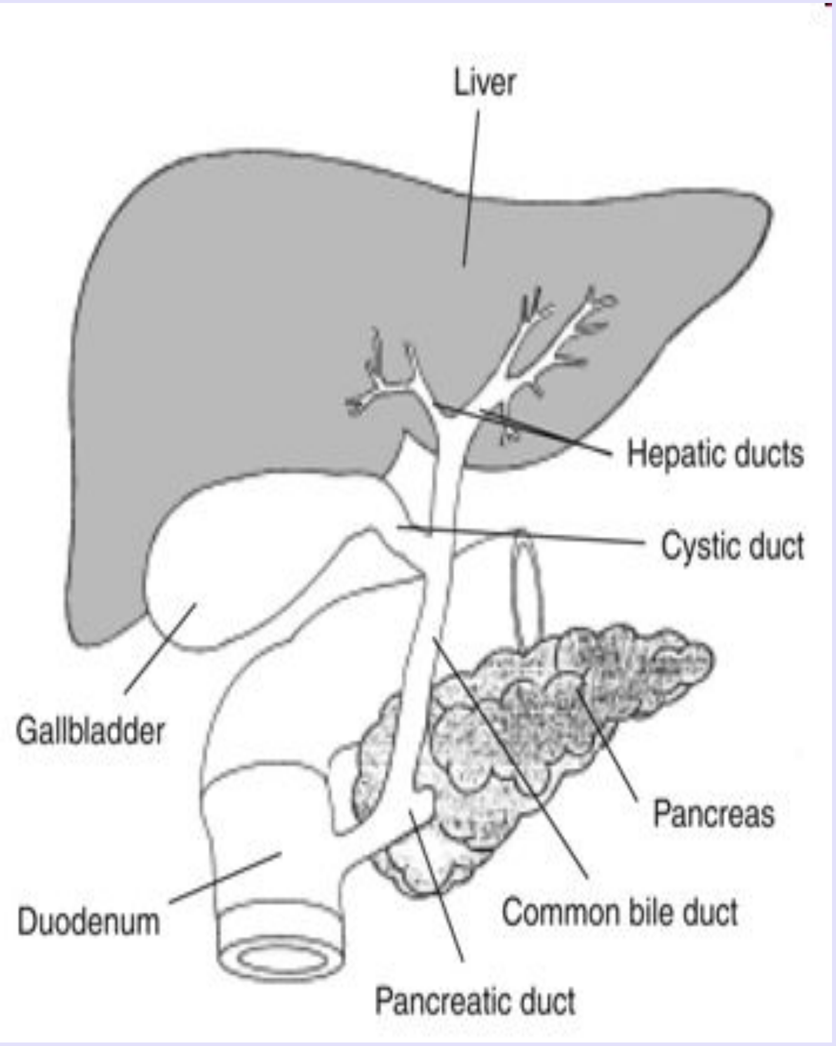
Absorption

- Absorption is the movement of nutrients, water, and electrolytes from the lumen of the intestine into the blood.
- In the small intestine , digestion is completed and most absorption occurs.
- Small absorbable units that result from digestion, along with water, vitamins, and electrolytes , are transferred from the digestive tract lumen into the blood or lymph.



Digestive tract and accessory digestive organs

- Digestive system is made up of gastrointestinal tract (GI tract) and accessory organs, which help in the process of digestion and absorption.
- GI tract is a tubular structure extending from the mouth up to anus, with a length of about 30 feet. GI tract is formed by two types of organs: 1. Primary digestive organs. 2. Accessory digestive organs.
- Accessory Digestive Organs
  Accessory digestive organs are those which help primary digestive organs in the process of digestion. Accessory digestive organs are: i. Teeth ii. Tongue iii. Salivary glands iv. Exocrine part of pancreas v. Liver vi. Gallbladder.



Digestive tract wall has four layers

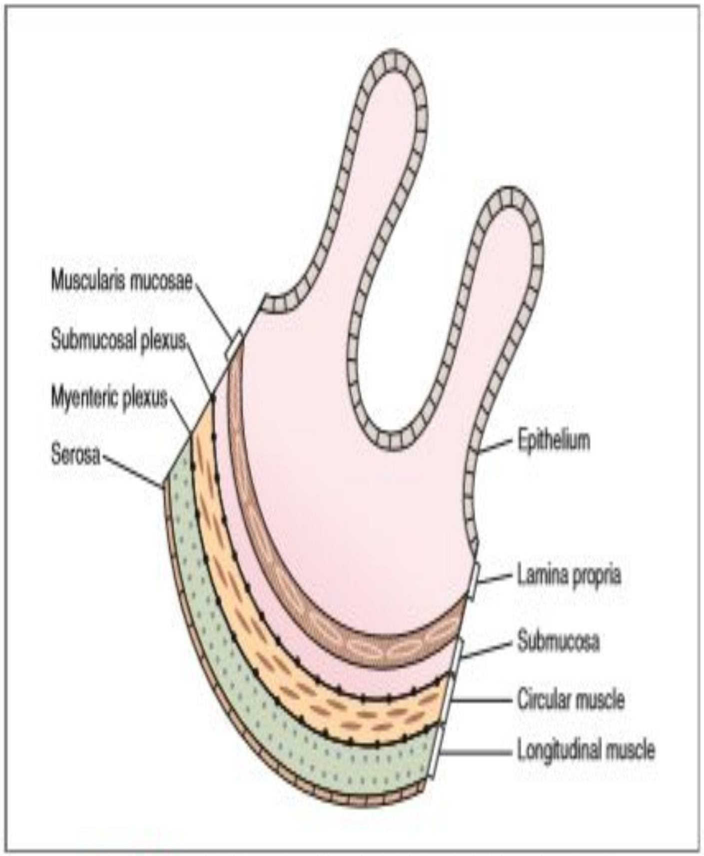


Figure 8-1 The structure of the wall of the gastrointestinal tract.

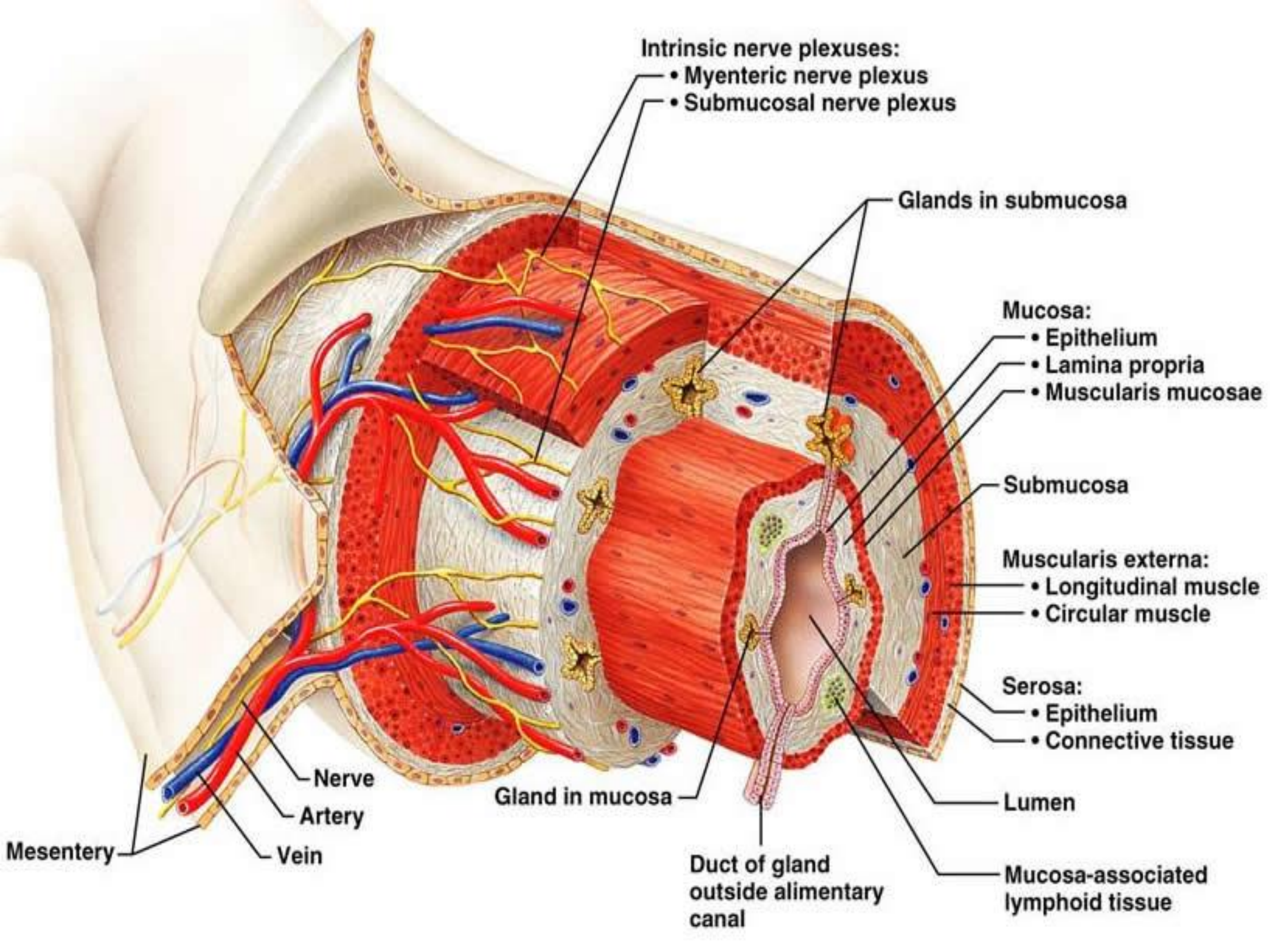
- The wall of the gastrointestinal tract has two surfaces, mucosal and serosal. The mucosal surface faces the lumen, and the serosal surface faces the blood. The layers of the gastrointestinal wall are as follows, starting from the lumen and moving toward the blood :

The Mucosa.

The submucosa,

The Muscularis Externa, and

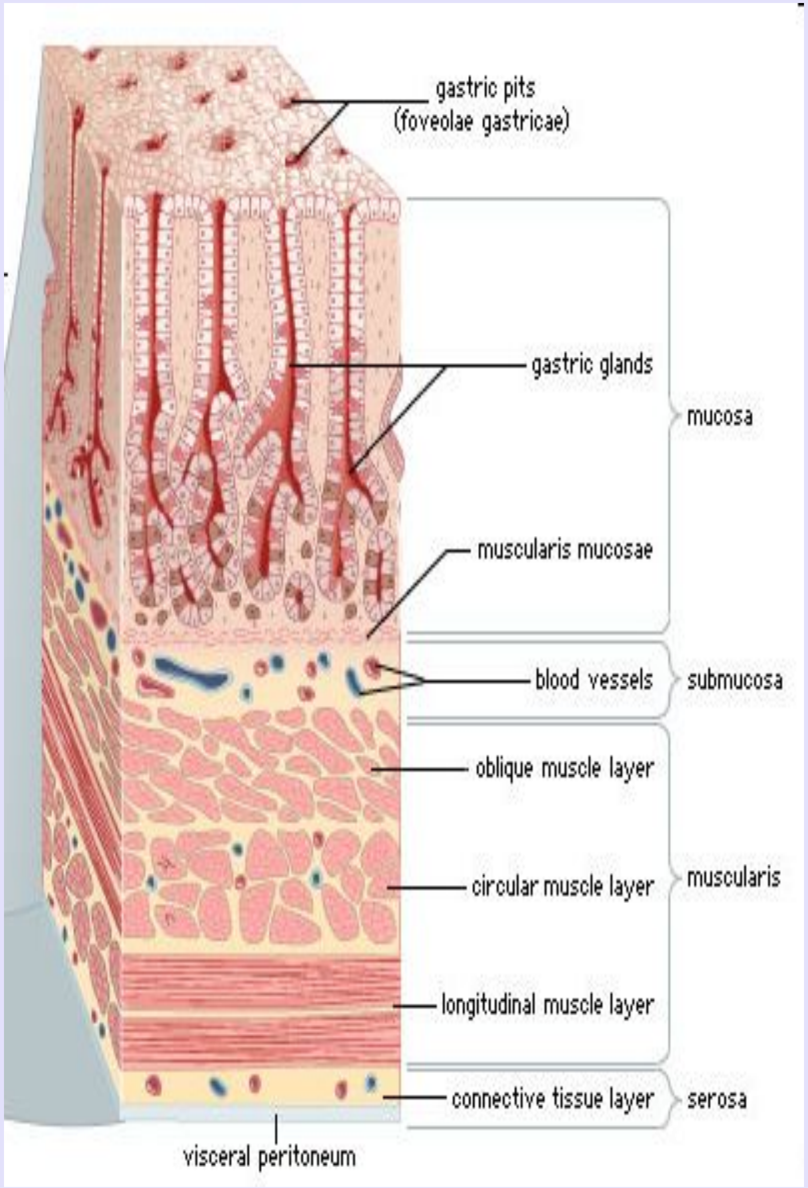
The serosa.


Intrinsic nerve plexuses:
• Myenteric nerve plexus
• Submucosal nerve plexus
Glands in submucosa
Mucosa:
• Epithelium
• Lamina propria
• Muscularis mucosae
Submucosa
Muscularis externa:
• Longitudinal muscle
• Circular muscle
Serosa:
• Epithelium
• Connective tissue
Lumen
Mucosa-associated lymphoid tissue
Duct of gland outside alimentary canal
Gland in mucosa
Nerve
Artery
Vein
Mesentery

Mucosa

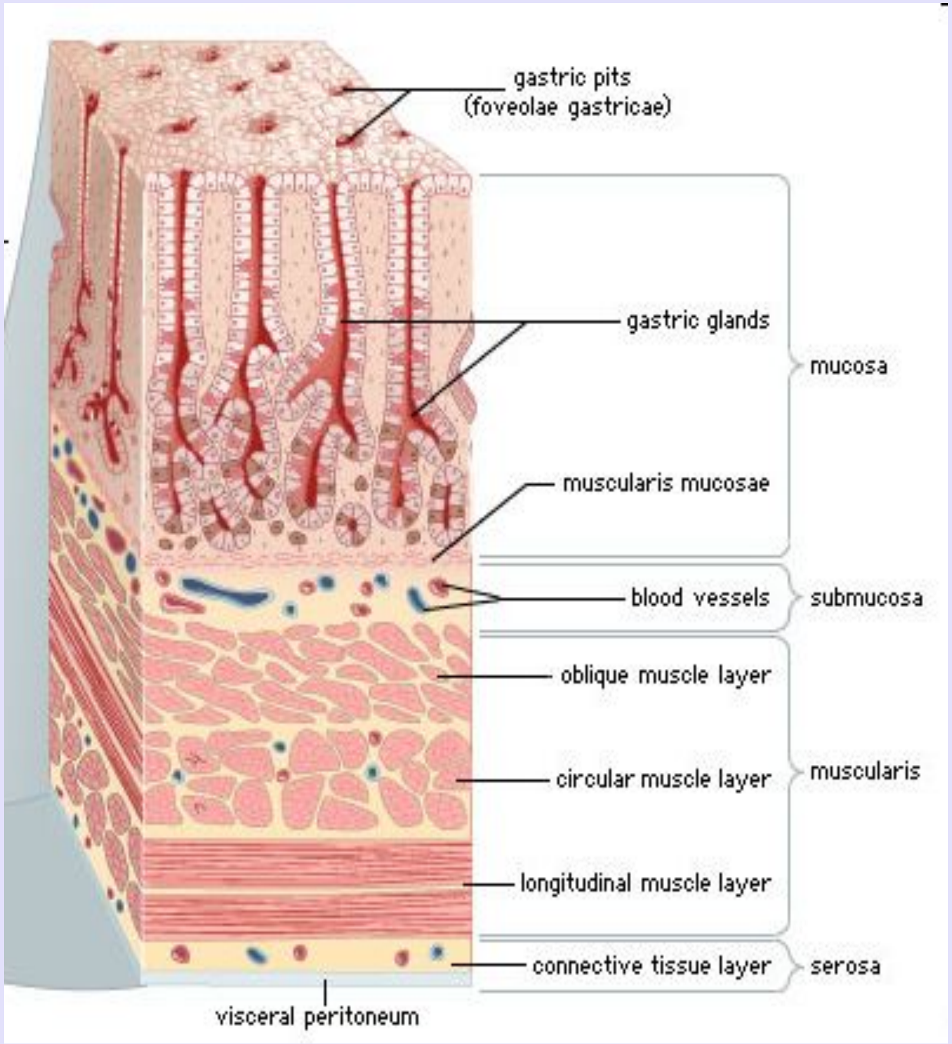
- The mucosal layer consists of a layer of epithelial cells, a lamina propria, and a muscularis mucosa (outermost smooth muscle layer).
- The epithelial cells are specialized to carry out absorptive and secretory functions.
- The lamina propria consists primarily of connective tissue, but it also includes blood and lymph vessels.
- The muscularis mucosae consists of smooth muscle cells; contraction of the muscularis mucosae changes the shape and surface area of the epithelial cell layer.
- mucosa lines the luminal surface of the digestive tract.

Lamina propria houses the gut associated lymphoid tissue **(GALT)**



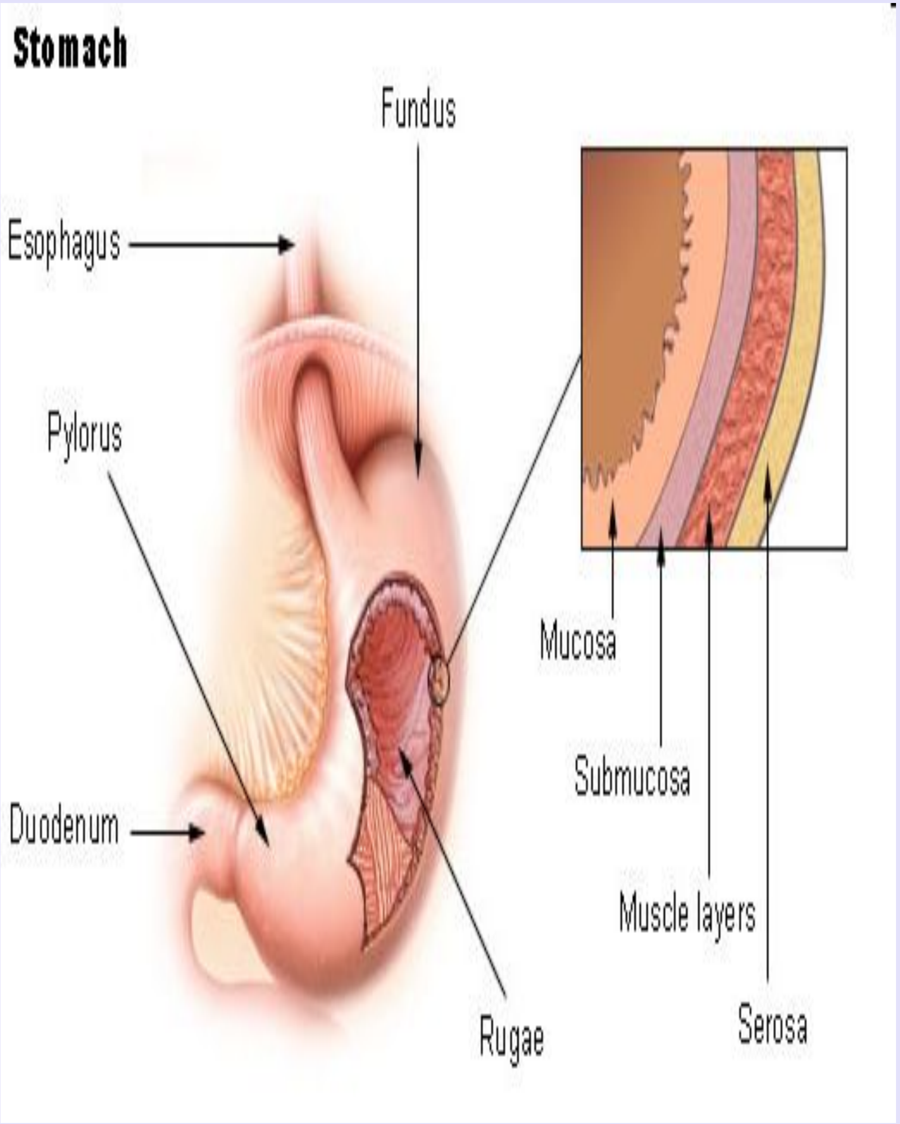
Submucosa

- Beneath the mucosal layer is a submucosal layer, which consists of collagen, elastin, glands, and the blood vessels of the gastrointestinal tract.
- A thick layer of connective tissue provides the digestive tract with its distensibility and elasticity.
- Submucosal plexus lies within the submucosa.



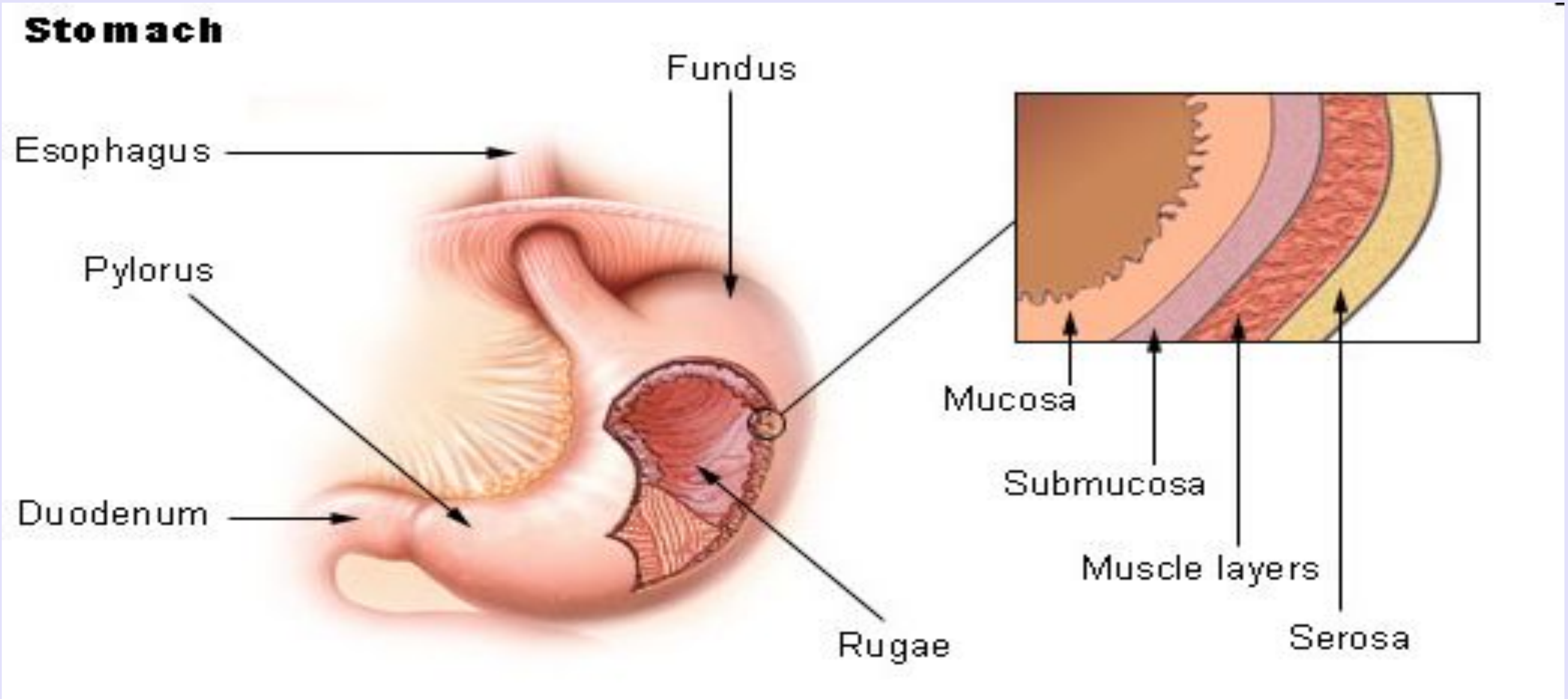
Muscularis externa

- Motility of the GIT is provided by two layers of smooth muscle, circular muscle and longitudinal muscle, which are interposed between the submucosa and the serosa.
- The longitudinal muscle layer is thin and contains few nerve fibers, whereas the circular muscle layer is thick and more densely innervated.
- Two plexuses, the **submucosal plexus** and the **myenteric plexus**, contain the nervous system of the gastrointestinal tract.
- Contractile activity of these smooth muscle layers produces the propulsive and mixing movements. The submucosal plexus (Meissner’s plexus) lies be tween the submucosa and the circular muscle.
- The myenteric plexus lies between the two muscle layers.



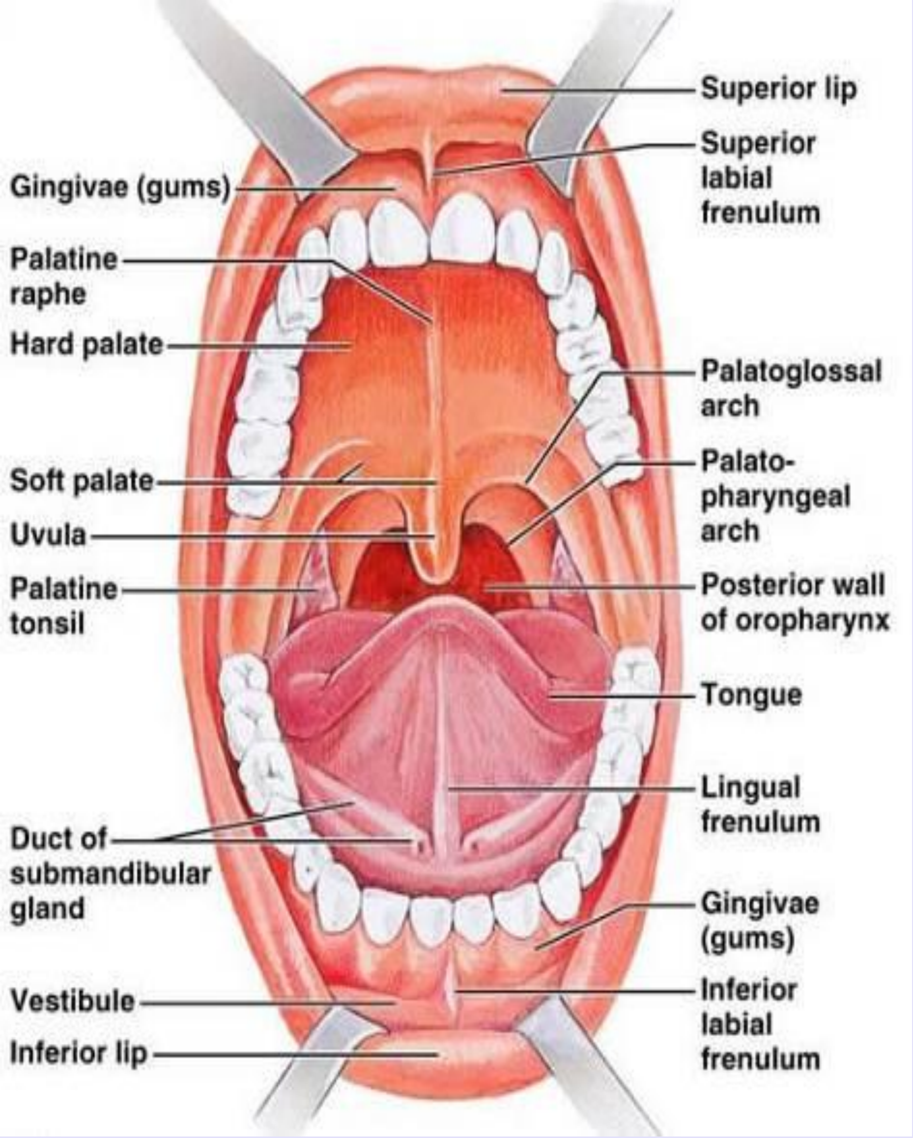
Serosa

- Outer connective tissue covering of the digestive tract is the serosa.
- Secretes a watery serous fluid that lubricates and prevents friction between the digestive organs and the surrounding viscera.



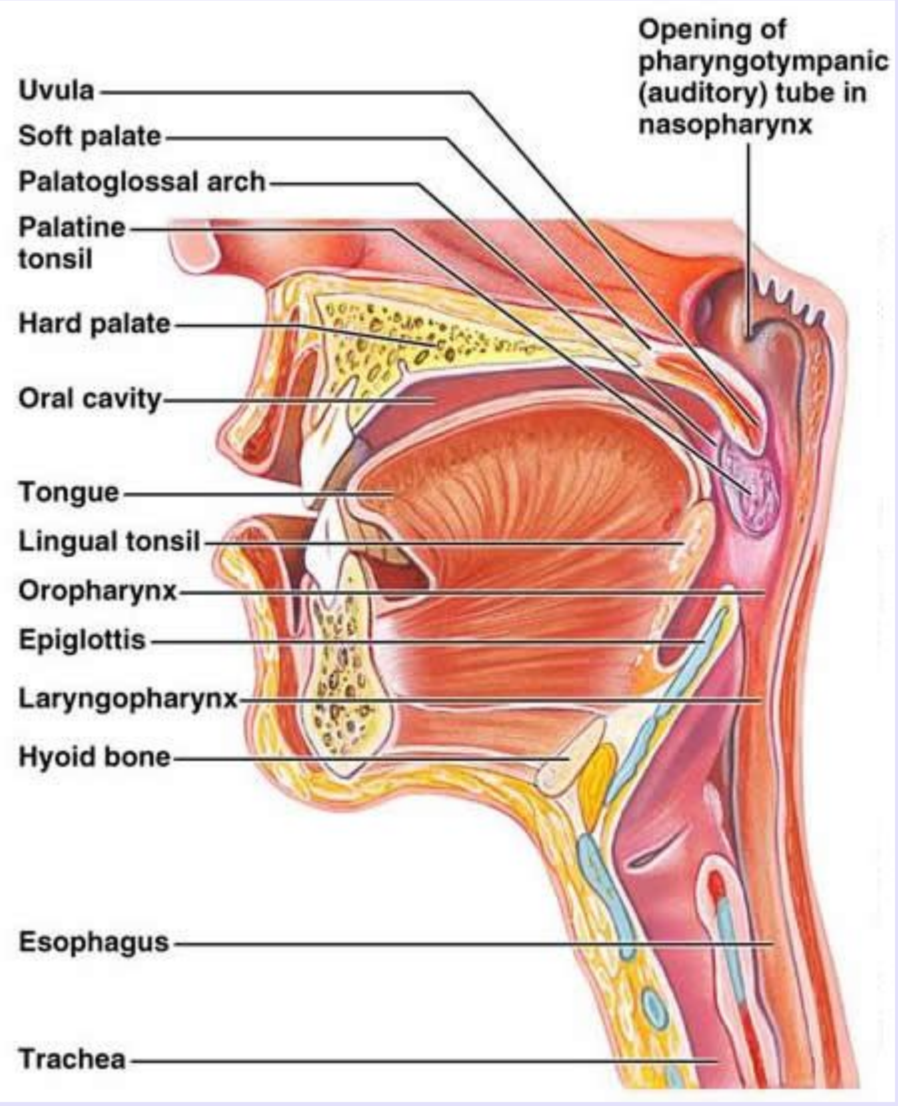
The oral cavity or mouth

- The **oral cavity or mouth** is the entrance to the digestive tract.
- Opening is formed by **muscular lips** , which help procure, guide and contain the food in the mouth.
- **Palate** allows breathing and chewing or sucking to take place simultaneously.
- **Uvula** seals off the nasal passages during swallowing.



The oral cavity or mouth

- **<u>Tongue :</u>** Movements of the tongue are important in guiding food within the mouth during chewing and swallowing.
- Major **<u>tastebuds</u>** are embedded in tongue.
- Pharynx acts as a common passage way for both the digestive system and the respiratory system.
- **<u>Tonsils</u>** are part of the body's defense mechanisms.

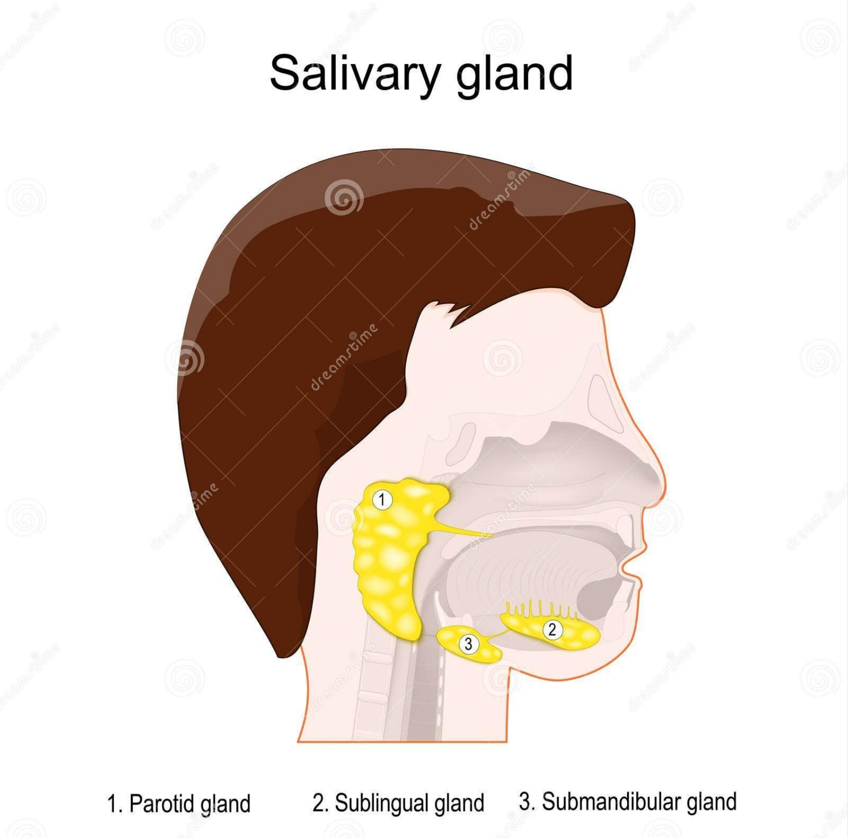


Teeth responsible for chewing

- First step in the digestive process is mastication or chewing.
- Chewing has three functions: (1) It mixes food with saliva, lubricating it to facilitate swallowing; (2) it reduces the size of food particles, which facilitates swallowing; and (3) it mixes ingested carbohydrates with salivary amylase to begin carbohydrate digestion.
- Chewing has both voluntary and involuntary components. The involuntary component involves reflexes initiated by food in the mouth. Sensory information is relayed from mechanoreceptors in the mouth to the brain stem, which orchestrates a reflex oscillatory pattern of activity to the muscles involved in chewing.

Saliva

- Saliva begins carbohydrate digestion.
- Is very important in oral hygiene, and
- Facilitates speech.
- Serves to lubricate the food bolus (aided by mucins).
- Hypotonic compared with plasma and alkaline.

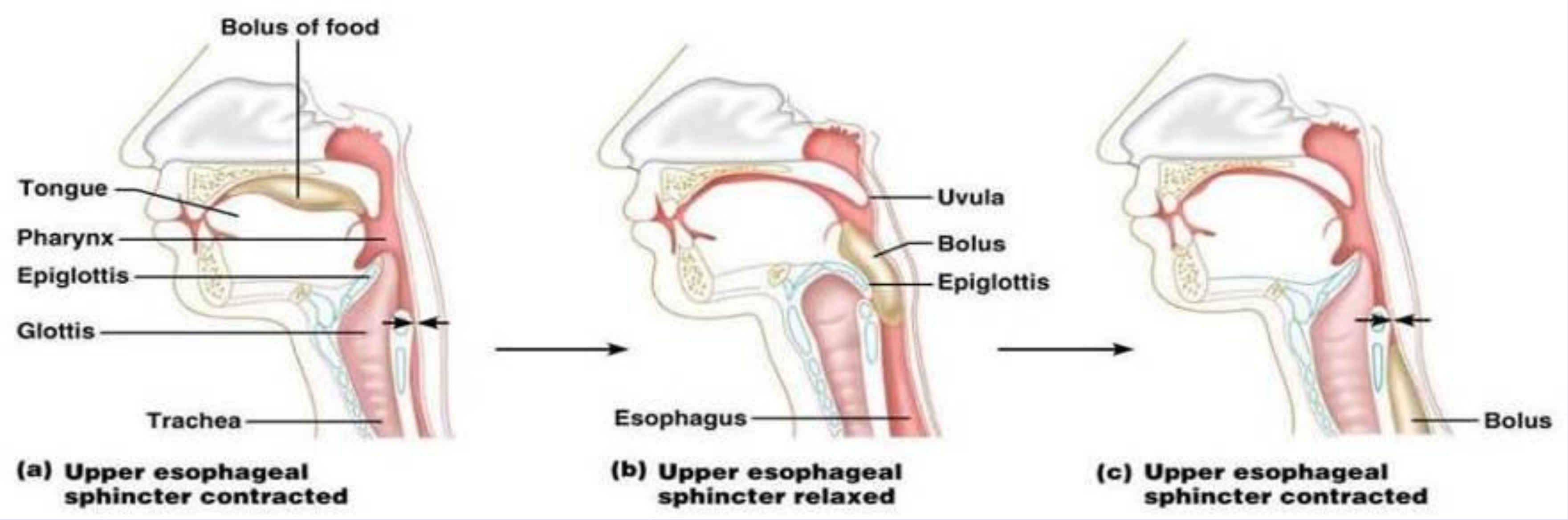


- Digestion in the mouth is minimal
- No absorption of foodstuffs occurs from mouth.

- **Imp:**
- **Some therapeutic agents can be absorbed by the oral mucosa.**
- **Prime example being a vasodilator drug, nitroglycerin to relieve angina attacks.**

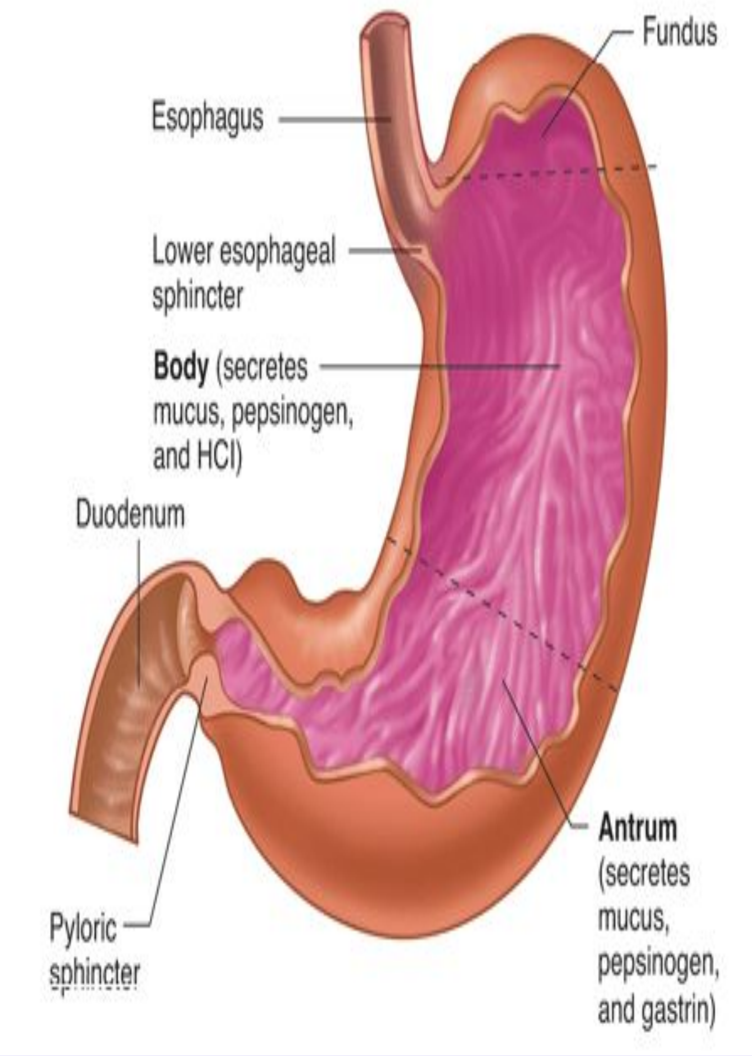
Pharynx and oesophagus

- The motility associated with the pharynx and oesophagus is **SWALLOWING.**
- During the oropharyngeal stage of swallowing , food is prevented from entering the wrong passageways.
- Peristaltic waves push food through the oesophagus.
- Gastroesophageal sphincter prevents reflux of gastric contents.



Stomach

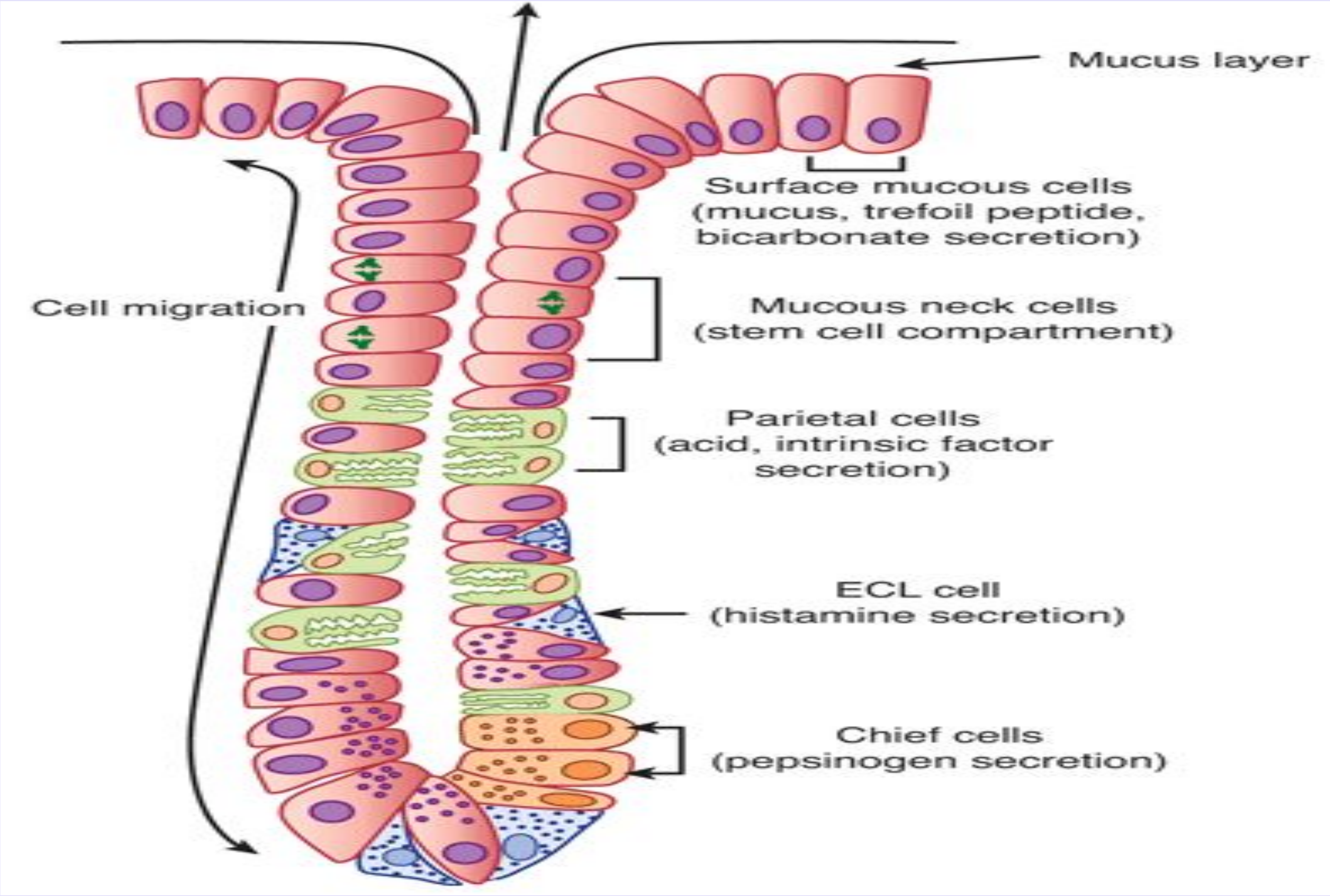
- Stomach is a hollow organ situated just below the diaphragm on the left side in the abdominal cavity. three anatomic divisions of the stomach: fundus, body, and antrum.
- On the basis of differences in motility, the stomach also can be divided into two regions, orad and caudad. The orad region is proximal, contains the fundus and the proximal portion of the body, and is thin walled.
- The caudad region is distal, contains the distal portion of the body and the antrum, and is thick walled to generate much stronger contractions than the orad region. Contractions of the caudad region mix the food and propel it into the small intestine.
- The gastric mucosa contains many deep glands. In the cardia and the pyloric region, the glands secrete mucus.
- In the body of the stomach, including the fundus, the glands also contain parietal (oxyntic) cells, which secrete hydrochloric acid and intrinsic factor, and chief (zymogen, peptic) cells, which secrete pepsinogens



Stomach

- These secretions mix with mucus secreted by the cells in the necks of the glands.
- Several of the glands open on a common chamber **(gastric pit)** that opens in turn on the surface of the mucosa.
- Mucus is also secreted along with HCO_3^- by mucus cells on the surface of the epithelium between glands.
- Carbohydrate digestion continues in the body of the stomach.
- Protein digestion begins in the antrum.
- No food or water is absorbed into the blood through the stomach mucosa.

Gastric pits



Gastric secretion

- The gastric mucosa secretes 1.2 to 1.5 litres of gastric juice per day.
- Gastric juice renders food particles soluble, initiates digestion (particularly of proteins), and converts the gastric contents to a semiliquid mass called chyme, thus preparing it for further digestion in the small intestine.
- Gastric juice is a variable mixture of water, hydrochloric acid, electrolytes (sodium, potassium, calcium, phosphate, sulfate, and bicarbonate), and organic substances (mucus, pepsins, and protein).

Small intestine

- Site where most digestion and absorption takes place.
- Consisting of duodenum, jejunum, and ileum.
- No further digestion is accomplished after luminal contents pass beyond the small intestine.
- No further absorption of nutrient occurs , although large intestine does absorb small amounts of salt and water.

Large intestine

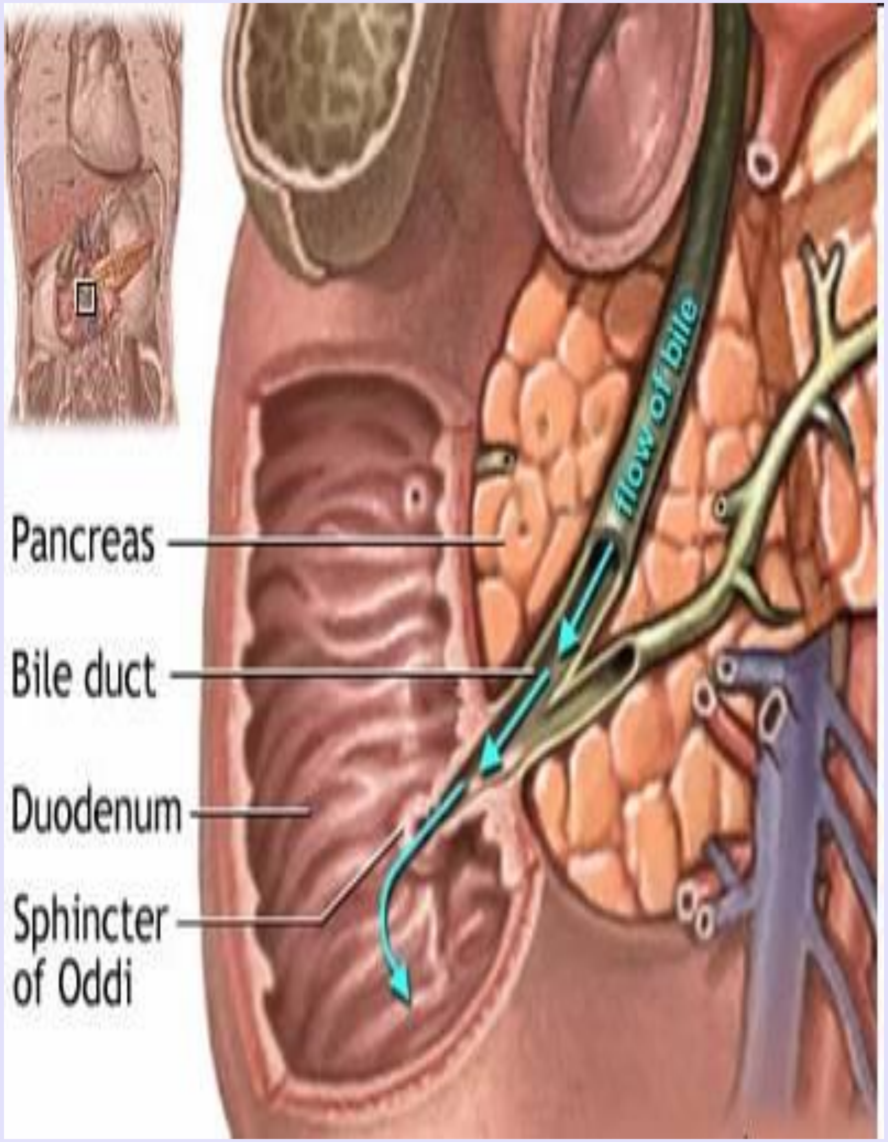
- Large intestine is primarily a drying and storage organ.
- Consists of colon, cecum, appendix and rectum.
- Colon normally receives about 500 ml of chyme from small intestine each day.
- Contents delivered to the colon consists of indigestible food residues (such as cellulose), unabsorbed biliary components and remaining fluid.

Overview of gastrointestinal hormones.

- **<u>Gastrin:</u>** Stimulates secretion by the parietal and chief cells.
- **<u>Secretin</u>**.and gastric secretion. Inhibits gastric emptying
- **<u>Cholecystokinin (CCK):</u>** inhibits gastric emptying and gastric secretion. .
- **<u>Gastric Inhibitary peptide (GIP):</u>** promotes metabolic processing of the nutrients once they are absorbed.
- **There are other functions also performed by above hormones.**

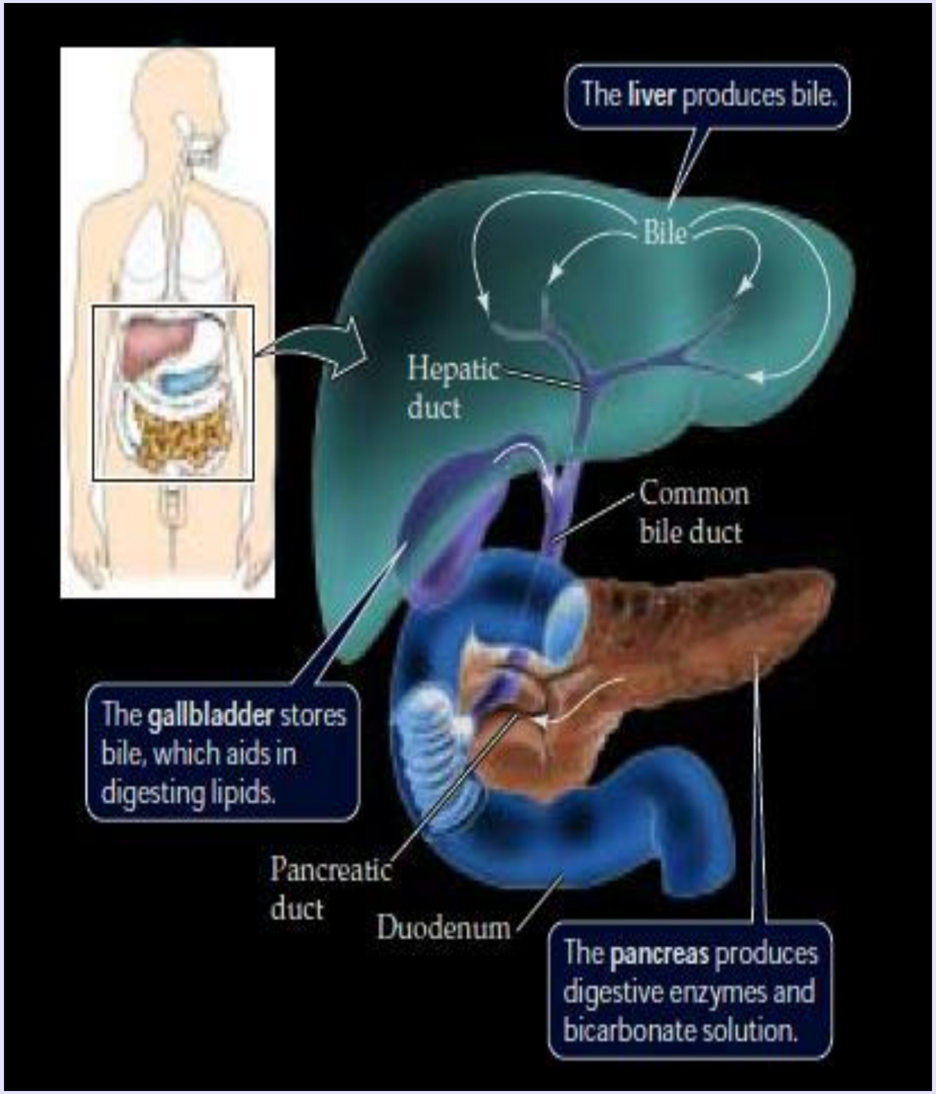
Biliary secretions

- The bile acids contained therein are important in the digestion and absorption of fats.
- Bile serves as a excretory fluid by which the body disposes of lipid soluble end products of metabolism.
- Bile is the only route by which the body can dispose of **CHOLESTEROL.**



Biliary secretions

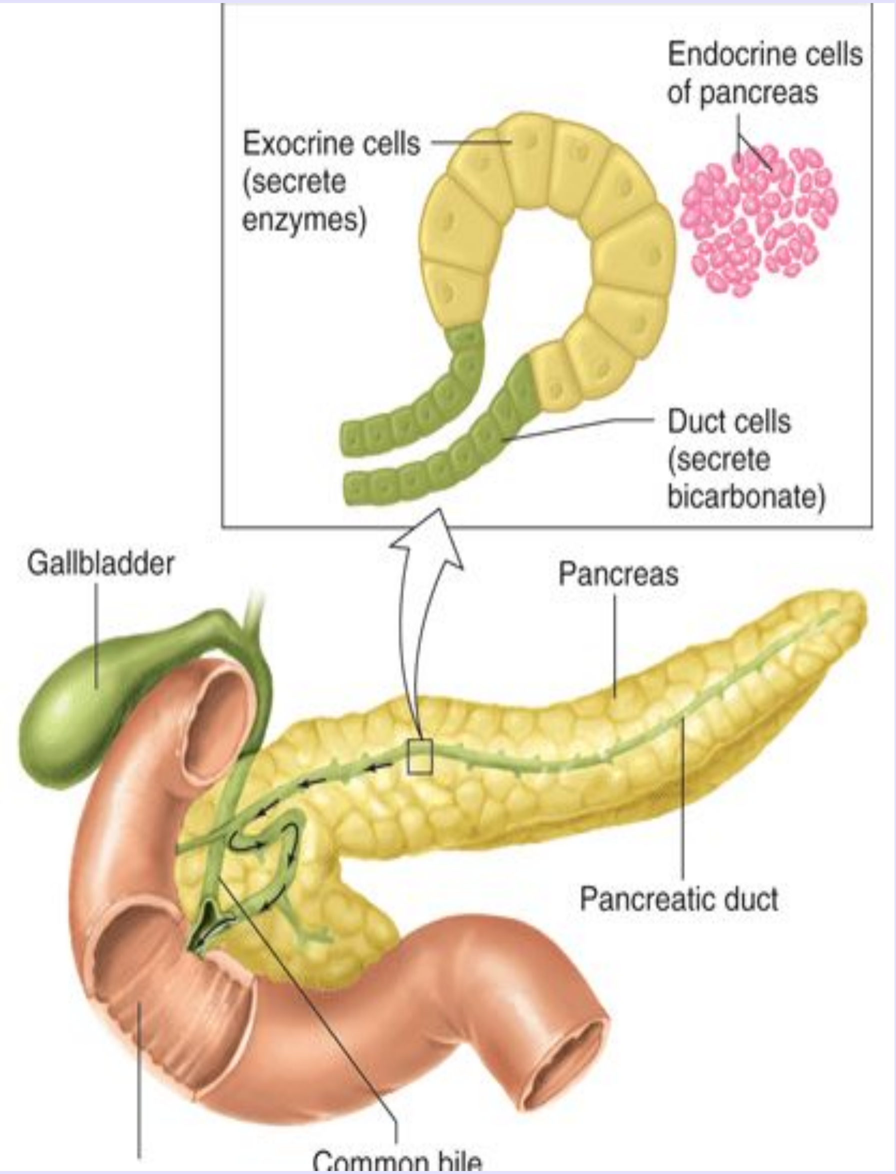
- Bile is made up of the bile acids, bile pigments, and other substances dissolved in an alkaline electrolyte solution .
- About 500 mL is secreted per day. Some of the components of the bile are reabsorbed in the intestine and then excreted again by the liver **(enterohepatic circulation).**

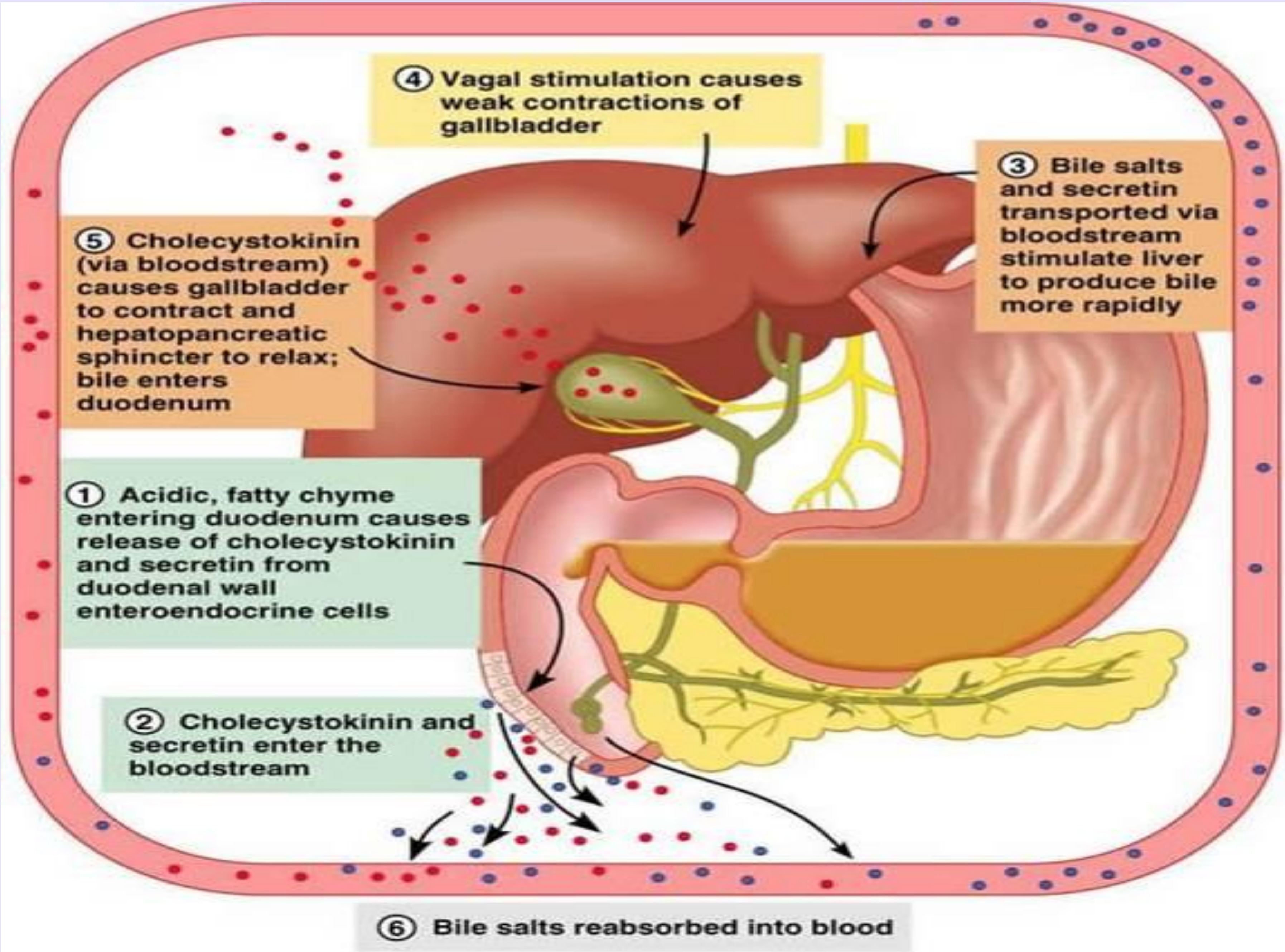


Pancreas

Anatomic Considerations:

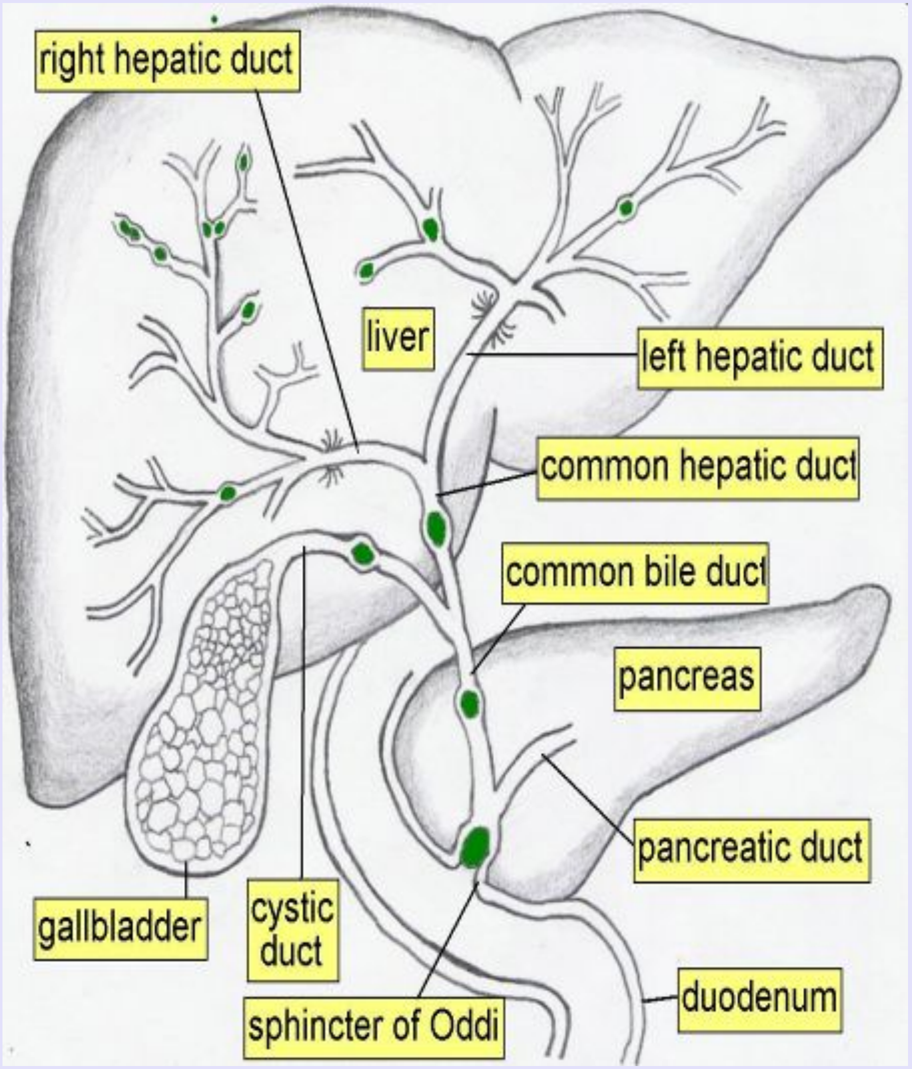
- The portion of the pancreas that secretes pancreatic juice is a compound alveolar gland resembling the salivary glands. Granules containing the digestive enzymes **(zymogen granules)** are formed in the cell and discharged by exocytosis




④ Vagal stimulation causes weak contractions of gallbladder
③ Bile salts and secretin transported via bloodstream stimulate liver to produce bile more rapidly
⑤ Cholecystokinin (via bloodstream) causes gallbladder to contract and hepatopancreatic sphincter to relax; bile enters duodenum
① Acidic, fatty chyme entering duodenum causes release of cholecystokinin and secretin from duodenal wall enteroendocrine cells
② Cholecystokinin and secretin enter the bloodstream
⑥ Bile salts reabsorbed into blood

Pancreas

- The small duct radicles coalesce into a single duct (**pancreatic duct of Wirsung),** which usually joins the common bile duct to form the ampulla of Vater.
- The ampulla opens through the duodenal papilla, and its orifice is encircled by the **sphincter of Oddi.**



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THANK
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